Designing and selection of lineage-specific baits for the genomic study of non-model organisms

T. R. Galise*, D. Cafasso*, P.M. Schlüter**, P Milet-Pinheiro***, M. Ayasse***, S. Cozzolino*

* Department of Biology, University of Naples Federico II

** Institute of Biology, University of Hohenheim

*** Institute of Evolutionary Ecology and Conservation Genomics, University of Ulm

Background and Aims

In non-model organisms, without a reference genome, genome-subsampling methods are often problematic because they are not targeted, producing randomly sampled sequences and acquisition of incorrect and missing data (1). In contrast, Hybrid capture-based target enrichment is often used for enrichment specific genes, exons and / or other genomic regions of interest. This approach designs the targeted capture through building RNA / DNA baits. However, there are still few workflows for baits design in non-model organisms (2). Here we propose a generic and reproducible method for the construction of baits using target sequences from orchid genomes and transcriptomes.

Workflow

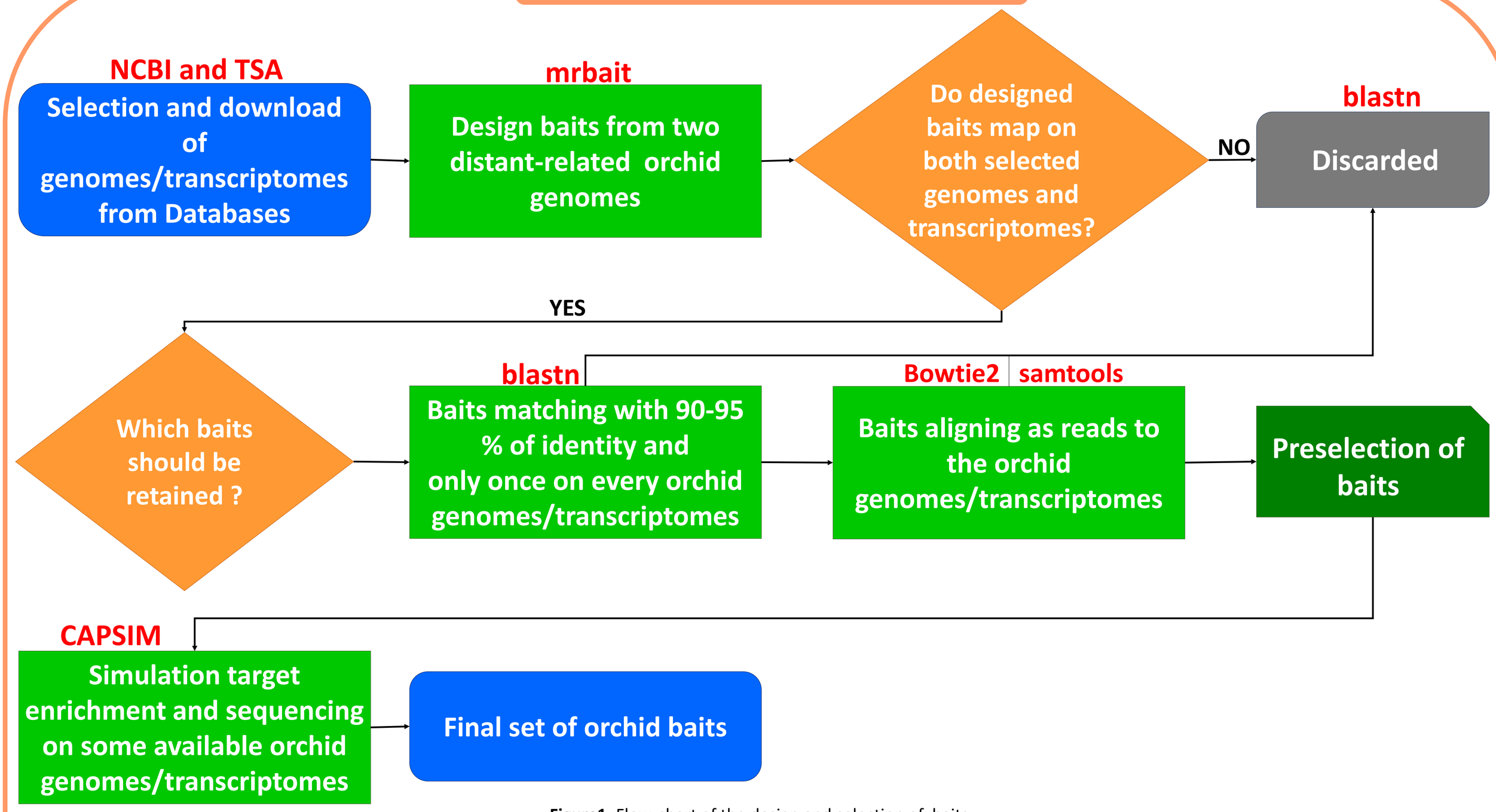

Figure1. Flow-chart of the design and selection of baits.

Data flow

- Baits design on genomes and transcriptomes of *Ophrys sphegodes* transcriptome (TSA: GHXJ01000001) and *Phalaenopsis equestris* (Genbank: GCA_001263595.1; TSA: GDHE01000001).
- In mrbait, 1,000bp and 500bp sliding windows are set. 1,680,457 baits were designed (769,809 Ophrys baits; 910,648 Phalaenopsis baits) with a length of 100bp and CG content between 20-50%
- For excluding baits too variable (<90 %) or too conserved (>95%) Ophrys baits within 90-95% identity interval with Phalaenopsis genome and transcriptome, and viceversa, were filtered by blastn retaining only those mapping once.
- The resulting baits (13894 from Ophrys, 11907 from Phalaenopsis) were further tested for mapping on alternative genome and transcriptome through simulations with Bowtie2/samtools and 23.200 baits were finally selected for synthesis.
- CAPSIM was used to simulate the dynamics of the capture process among the selected baits and some available orchid genomes/transcriptomes by defining an Illumina sequencing of 150bp single-end reads. The simulation generates approx. 17,000 raw reads resulting from cross-hybridization with selected baits (**Figure2**).

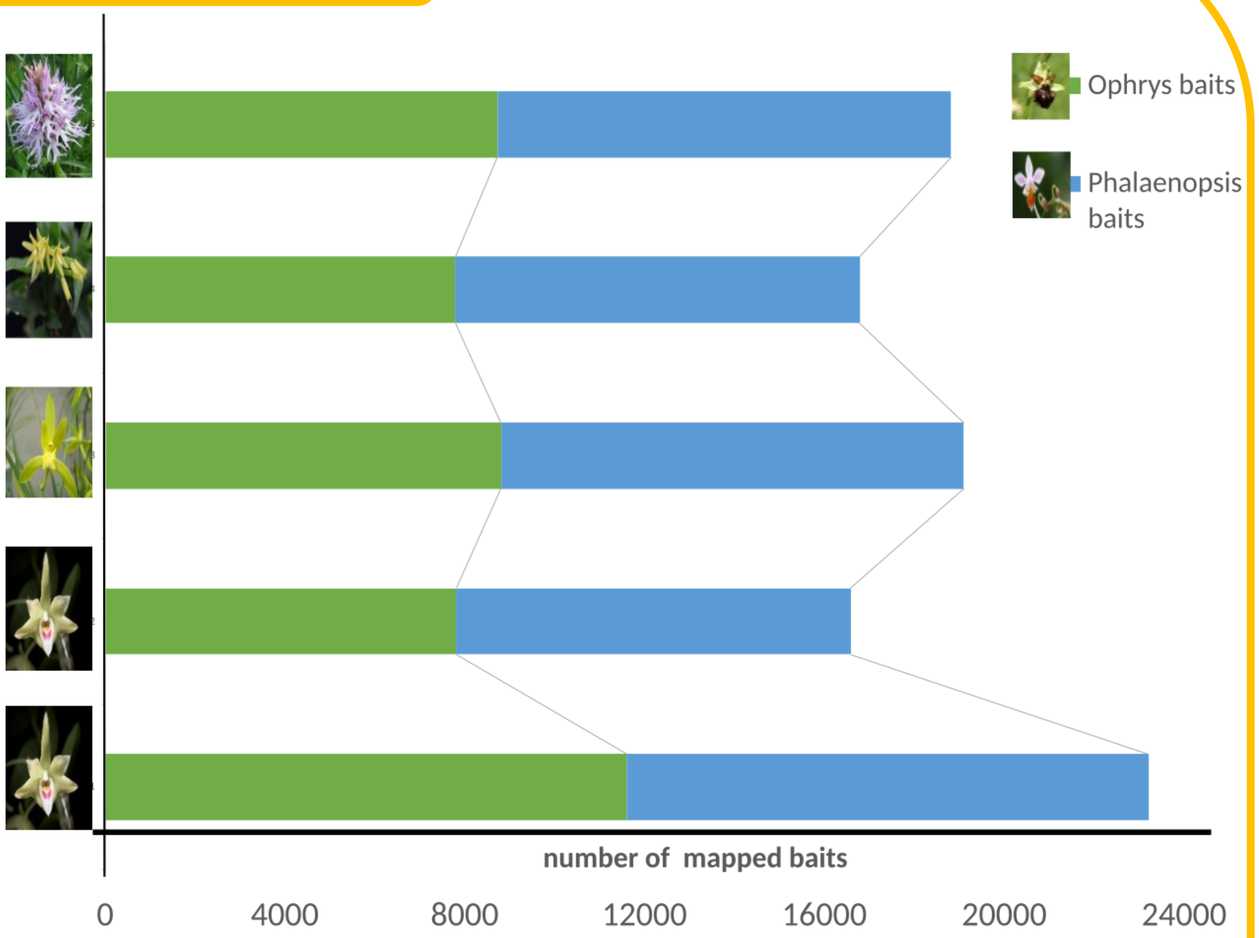

Figure2. CAPSIM simulation of selected 23.200 orchid baits (designed from *Ophrys* and from *Phalaenopsis*) hybridizing on the *Orchis italica* trascriptome (3), *Apostasia shenzhenica* genome (Genbank: GCA_002786265.1), *Cymbidium faberi* trascriptome (TSA: GDHD01000001) *and Dendrobium catenatum* genome (Genbank: GCA_001605985.2) and trascriptome (TSA: GEZV01000001).

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2. Bossert S, Danforth BN. On the universality of target-enrichment baits for phylogenomic research. Methods Ecol Evol. 2018;9(6):1453–60. doi:10.1111/2041-210X.12988
3. De Paolo S, Salvemini M, Gaudio L, Aceto S. De novo transcriptome assembly from inflorescence of Orchis italica: Analysis of coding and non-coding transcripts. PLoS One. 2014;9. doi:10.1371/journal.pone.0102155